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COMMENTARY

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TYPE 2 DIABETES MELLITUS IN PARASITIC INFECTIONS PATIENTS LIBYA

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Abstract :

Diabetes is becoming a major chronic disease worldwide. Little is known about the prevalence of intestinal parasitic infections among them. The purpose of this study was to investigate the prevalence of his type 2 diabetes in Benghazi, Libya. In a cross-sectional case-control study, 200 type 2 diabetic patients were recruited from Benghazi Diabetes Center, and 100 non-diabetic patients were recruited from Benghazi Medical Center and Elhya Clinic as a control group. Their ages ranged from 18 years old to 55 years old, and there were no cases of diarrhea between January and December 2016. Sociodemographic characteristics were recorded during the interview. Fresh stool samples were collected and processed, and the data were analyzed.

Keywords : Diabetes mellitus, Intestinal parasites, Type 2 diabetes

Introduction :

There are two main types of intestinal parasites: helminths and protozoa, which are important causes of infection in immunocompromised people. In healthy people, they usually cause less aggressive disease. This is a public health problem, especially in tropical countries where sanitation is poor and food and water sources are contaminated. Developing countries are more susceptible to intestinal and extra intestinal parasitic diseases. Even in well-developed areas, some of these parasites can cause serious disease in immunocompromised patients and young children. Approximately 340 parasites infect an estimated 3.5 billion people and cause clinical signs and symptoms in approximately 450 million. Weakness or impairment of the immune system, such as chronic internal diseases or metabolic disorders, can lead to an increased risk of infection, chronic infections, and intestinal parasites [1].

The prevalence of intestinal parasites was higher in diabetic patients (40.0%) than in non-diabetic patients (16.0%). Specifically, the following intestinal parasites were common in diabetic patients: Entamoeba coli (10.5%), Cryptosporidium parvum (8.5%). In the control group, Entamoeba histolytica (10%) and Ascaris Lumbricoides (3%) were the most frequently isolated parasites.

There were significant associations between parasite prevalence and diarrhea ($p = 0.001$), education ($p=0.000$), and single and multiple infections ($p = 0.001$).

The association between parasitic infections and gender and age was not significant in all samples ($p = 0.203$, $p = 0.56$). This study highlights the need to increase physician awareness of the prevalence of intestinal parasites in patients with type 2 diabetes [2].

Commentary:

Diabetes is one of the metabolic diseases that can affect the immune system. It has been reported that diabetic patients have a weakened immune system and are therefore more susceptible to many infections, including parasitic infections.

The prevalence of diabetic patients infected with intestinal parasites in our country is poorly recognized, and the effectiveness of treatment and prevention methods is unknown. Therefore, we conduct this study to determine the burden of parasitic infections in both diabetic and non-diabetic populations of the Benghazi population [3].

In a cross-sectional case-control study, 200 type 2 diabetic patients 134 women, 66 men were recruited from the Benghazi Diabetes Center, and 100 non-diabetic patients 73 women, 27 men) were recruited as a control group from the Benghazi Medical Center person was hired. At the Elhia in clinic, from January to December 2016, they examined patients from the age of 18 to the age of 55 with or without diarrhea. Sociodemographic characteristics were recorded during the interview. Fresh stool samples were collected and processed. Glycated hemoglobin (HbAlc) was measured to detect DM in the control group [4].

All participants in this study were interviewed using a questionnaire. Stool samples were collected in dry, clean, leak-proof plastic containers and immediately transported to the laboratory at the Benghazi Institute for Health Professions. Half of each sample was stored in her 10% formalin to prevent bacterial effects. These samples were examined macroscopically for appearance, color, viscosity, blood mucosa and presence of adults or larvae and microscopically by direct and formalin ether concentration methods [2-4]. Direct microscopy of swabs in saline (0.85% NaCl solution) confirmed the presence of motile intestinal parasites and trophozoites under a light microscope (40x magnification) and eggs using Lugol's iodine, detected larvae, trophozoites and cysts.

Intestinal parasites. Additionally, to increase the probability of detecting protozoan oocysts, an enrichment method was used in which stool samples were mixed with formalin, treated with ether, and then centrifuged. The ether, formalin, and debris layers were discarded and the residue was used to test for the presence of intestinal parasites. We also used a modified Ziehl-Neelsen staining technique to detect oocysts of the opportunistic coccidial intestinal parasites Cryptosporidium spp, Isospora belli and Cyclospora cayetanensis. Data were analyzed using the SPSS version 17 statistical package. Comparisons of parasite abundance between cases and controls were performed using the chi-square test [5].

The prevalence of parasitic infections in our study was 40% in diabetic patients, which was significantly higher than in nondiabetic patients (16%). This result is consistent with several previous studies. A study in Iran in 2015 found that the prevalence of parasitic infections in diabetic patients was 26.3%, higher than the control rate of 6.8%. We also agree with Olusegun et al [4]. This may be explained by the disruption of the immune response in diabetic patients. A study conducted in southeastern Turkey in 2001 showed a lower prevalence of parasitic infections in diabetic patients (10% and 47%) compared to non-diabetic patients (23.5% and 55.9%).

The researchers based their findings on the fact that diabetic patients visit the doctor more frequently than non-diabetic patients because they are more likely to be examined and treated for possible intestinal parasitic infections. Of note, both of these latter studies were conducted in areas known to have high parasite infection rates. On the other hand, we will conduct research in areas with low parasite infection rates. Therefore, the effects of diabetes may be attenuated in areas where the parasite is endemic [6,7]. The association between diabetes status and risk of parasitic infection was highly significant in our study (OR = 3.5) H. Diabetes increases the risk of parasitic infections. However, in contrast to the in 2001, it was shown to have a lower risk of diabetes. Several previous studies have shown no difference between gender and age, which is consistent with our results in all samples (p-value = 0.203) and in diabetic patients [3,6]. Many studies have shown a higher prevalence of parasitic infections in women and older adults, which is similar to our findings for non-diabetic patients (her P value for gender = 0.04, Her P value for age = 0.001) [4]. Therefore, diabetic status appears to moderate the effects of age and gender on the risk of parasitic infection. Lower education level and a history of diarrhea were strong predictors of parasitic infections, especially mixed infections, which is consistent with several previous studies [8].

Conflict of interest:

Author declares there is no conflict of interest.

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